Chapter 10 — Responsive Design

People open the same page on a 360px-wide phone, a tablet turned sideways, a 1440px laptop and a 2560px monitor — and some of them have set a larger default font, zoomed to 200%, switched their system to dark mode (light text on a dark background) or asked it to reduce motion (fewer animations, which can cause dizziness or nausea for some people). **Responsive design** is the practice of building *one* page that works well in all of those situations, instead of a separate "mobile site". Chapters 7–9 gave you the layout tools (flow, flexbox, grid); this chapter is about making those layouts **adapt**.

What you'll learn

#	Topic	You will be able to…
10.1	What responsive design is	name its three ingredients and explain mobile-first (design for phones first, then add for bigger screens)
10.2	The viewport meta tag and viewport units	explain why phones need it, and choose between `vh`, `svh`, `lvh`, `dvh`
10.3	Media queries	write queries with types, features, range syntax and logic, and use them from JavaScript
10.4	Mobile-first in practice	build a page that goes from 1 to 3 columns with two breakpoints (screen widths at which the layout changes)
10.5	User-preference queries	support dark mode, higher contrast and reduced motion
10.6	Container queries	make a component adapt to the space *it* is given, not the screen
10.7	Fluid typography with `clamp()`	derive a fluid font-size formula and keep it accessible
10.8	Responsive images recap	pick the right image with `srcset` / `sizes` / `<picture>` and `image-set()`
10.9	Intrinsic layouts	build layouts that adapt with no breakpoints at all
10.10	Testing responsive design	test with DevTools (the browser's built-in developer tools), real devices, zoom — and automate an overflow check

Every example is a page in `code/`. The screenshots in `img/` were taken in headless Chrome (Chrome driven by a script, with no window) by `code/make_figures.mjs`, and the `NN_*.mjs` scripts print what the browser actually computed (which queries match, resolved grid columns, font sizes). They use the helper `code/measure.mjs` — the same idea as in Chapters 7–9, extended with **viewport width**, **phone emulation** (making desktop Chrome behave like a phone) and **OS-preference emulation** (pretending the operating system has settings such as dark mode or high contrast). Many screenshots are "contact sheets": the *same* page at several widths, side by side, with the width in red above each one.

10.1 What responsive design is

What it is

A **responsive** page rearranges and resizes itself to fit the **viewport** (the visible area of the browser window) and the user's settings. The term was coined by Ethan Marcotte in 2010, with three ingredients:

Ingredient	Meaning	Today's tools
Fluid grids	Widths in relative units (%, `fr`, `vw`) instead of fixed pixels	Flexbox, Grid, `min()` / `max()` / `clamp()`
Flexible media	Images and videos shrink with their container	`max-width: 100%; height: auto`, `srcset`, `aspect-ratio`
Media queries	CSS rules that apply only under conditions (e.g. "at least 640px wide")	`@media`, and now `@container`

Analogy: water takes the shape of whatever glass you pour it into. A fixed-width page is an ice cube — it fits only one glass.

Why it matters

- Most web traffic worldwide is from phones; a page that needs sideways scrolling or pinch-zoom (spreading two fingers to enlarge) on a phone loses readers.
- Search engines index the mobile version of a page first ("mobile-first indexing" at Google), so a broken mobile layout hurts discovery (Chapter 14).
- Accessibility law and WCAG 2.2 (the Web Content Accessibility Guidelines) require content to **reflow** (rewrap into a narrow column without sideways scrolling) at 320 CSS pixels wide and survive 200% text zoom (§10.10, Chapter 3).

Mobile-first vs desktop-first

	Mobile-first	Desktop-first
Base CSS (no query)	The phone layout	The desktop layout
Queries use	`min-width` / `width >=` — *add* features as space grows	`max-width` / `width <` — *undo* features as space shrinks
Small devices download/parse (read and process)	only the rules they need, plus harmless overrides	desktop rules, then overrides to undo them
Design pressure	Forces you to decide what matters most first	Tends to cram a big design into a small screen

Mobile-first is the common default today because the single-column phone layout is the simplest — normal flow already stacks things — and every larger layout is an *enhancement*. That is **progressive enhancement**: start with something that works everywhere, then add.

Responsive vs adaptive

An **adaptive** site serves (or switches between) a few fixed-width layouts; a **responsive** one is fluid *between* breakpoints. Modern practice is responsive, often with fewer breakpoints thanks to intrinsic layouts (10.9) and container queries (10.6).

Self-check

1. Name the three ingredients of responsive design. → Fluid grids, flexible media, media queries.
2. In mobile-first CSS, which kind of width query do you mostly write? → `min-width` (or `width >= …`).

10.2 The viewport meta tag and viewport units

What it is

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

This line in `<head>` (Chapter 2) tells a mobile browser: "lay the page out at the device's width in CSS pixels, at 100% zoom".

The problem it solves

When the iPhone launched (2007), almost every site was designed for ~1000px-wide desktops. So mobile browsers **pretend** to be wide: without the tag they lay the page out in a **layout viewport** of about **980px**, then zoom out so it fits the screen. Old sites looked like tiny thumbnails you could pinch-zoom — but a responsive page would never see its "phone" media queries fire, because the browser claims it is 980px wide.

A **CSS pixel** is not a hardware pixel. A phone with a **device pixel ratio** (DPR) of 3 draws each CSS pixel with 3×3 physical pixels; `width=device-width` means "as many CSS pixels as the screen is wide", e.g. 390 on a typical modern phone.

Proof

`code/01_no_viewport.html` and `code/01_viewport.html` are the same page except for the tag. `code/01_viewport.mjs` opens both on an emulated 400px phone (Puppeteer — a Node.js library that controls Chrome — has an `isMobile: true` option that makes Chrome honour the meta tag like a real phone):

```
01_no_viewport.html  layout 980px, visible 980px, scale 0.408, font-size 16px
01_viewport.html     layout 400px, visible 400px, scale 1, font-size 16px
```

Without the tag, the page is laid out at 980px and scaled by 400 / 980 ≈ 0.408 — so the "16px" text is really about **6.5px** on screen.

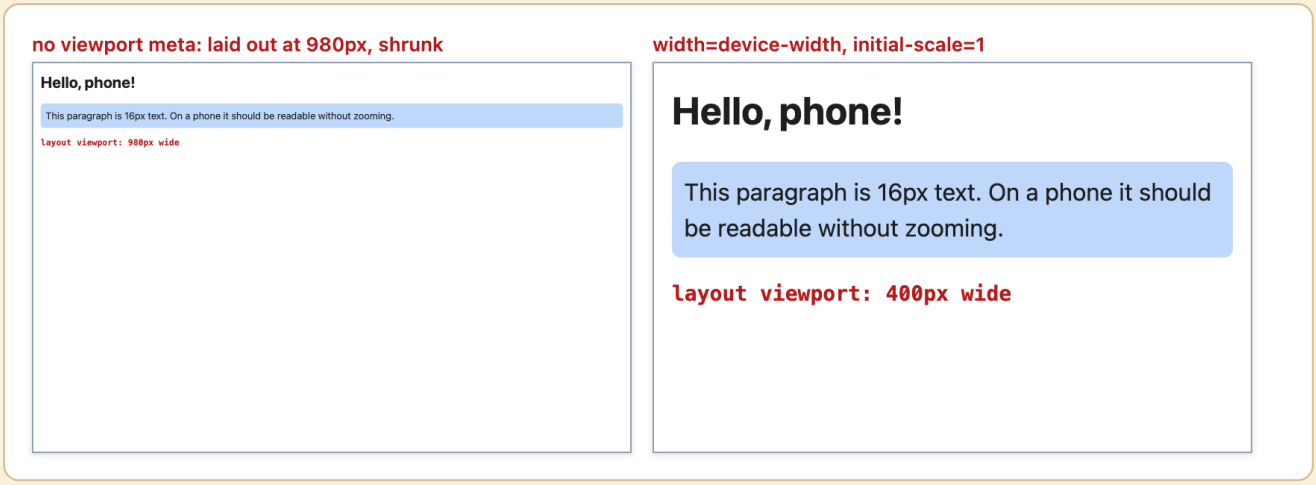


Figure 10.1 — The same page on a 400px phone, without and with the viewport meta tag.

The values

Key	Typical value	Notes
`width`	`device-width`	Layout viewport = screen width in CSS px. A number (`width=600`) is possible but rarely right.
`initial-scale`	`1`	Start at 100% zoom. Also stops iOS from changing zoom on rotation.
`maximum-scale`, `user-scalable=no`	**don't**	Blocking zoom fails WCAG 1.4.4 (Resize text). iOS Safari has ignored `user-scalable=no` since iOS 10 for this reason.
`viewport-fit`	`cover`	For notched phones (screens with a camera cut-out); then use `env(safe-area-inset-*)` padding (values the browser supplies for the unsafe edges).

Viewport units

`vw` and `vh` are 1% of the viewport's width and height (Chapter 6 introduced the units). On phones the viewport height **changes** as the address bar slides in and out, so CSS Values 4 added three variants

(Baseline since December 2022):

Unit	1% of…	Behaviour on a phone
`svh` / `svw` (small)	the viewport with browser toolbars **shown**	never too tall; may leave a gap when toolbars hide
`lvh` / `lvw` (large)	the viewport with toolbars **hidden**	may be hidden behind toolbars at first
`dvh` / `dvw` (dynamic)	the viewport **right now**	follows the toolbars; can cause relayout (the browser recalculating the layout) while scrolling
`vh` / `vw` (classic)	in practice the **large** viewport on mobile	why `height: 100vh` "overflows" on phones

Rules of thumb: `min-height: 100svh` or `100dvh` for full-screen heroes (the large banner section at the top of a page) and the sticky footer of §8.7; `vw` for fluid type (10.7), but never alone — it ignores zoom settings. On desktop all four heights are equal. Also available: `vi`/`vb` (inline/block direction) and `vmin`/`vmax`.

Common mistakes

- Forgetting the tag: every phone breakpoint silently fails.
- Disabling zoom "to feel like an app" — an accessibility failure.
- `width: 100vw` on a page with a vertical scrollbar: on desktop `100vw` *includes* the scrollbar, so the element is ~15px too wide and causes sideways scrolling. Use `width: 100%`.

Self-check

1. Without the tag, how wide does a phone lay the page out? → About 980 CSS px (then zooms out).
2. Which unit should a full-height mobile hero use to avoid being covered by the address bar? → `svh` (always fits) or `dvh` (tracks the toolbar).
3. Why avoid `100vw` for full-width elements? → It includes the scrollbar width on desktop and causes horizontal overflow.

10.3 Media queries

What it is

A **media query** is a condition about the device or environment. Rules inside `@media (condition) { … }` apply only while the condition is true, and the browser re-evaluates it live (e.g. when you rotate a phone or resize a window).

```
@media screen and (width >= 40em) { .cards { grid-template-columns: repeat(2, 1fr); } }
```

Media queries don't add specificity (the score that decides which competing CSS rule wins); inside them normal cascade (CSS's conflict-resolution) rules apply (Chapter 6) — which is why mobile-first queries are written **after** the base rules they override.

Where you can use them

Place	Example
CSS at-rule (a CSS statement starting with `@`)	`@media (width >= 40em) { … }`
`<link>`	`<link rel="stylesheet" href="print.css" media="print">` — still downloaded, but not render-blocking (it doesn't delay the first paint) when it doesn't match
`@import`	`@import url(wide.css) (width >= 64em);`
`<source>` in `<picture>`	`<source media="(width >= 800px)" srcset="wide.jpg">` (Chapter 4)
JavaScript	`matchMedia('(width >= 64em)')`

Anatomy

1. **Media type** (optional): `all` (default), `screen`, `print`. Other types (`tv`, `handheld`…) are deprecated (officially retired; don't use them).
2. **Media features** in parentheses: `(width >= 40em)`, `(orientation: landscape)`, `(hover: hover)`.
3. **Logic**: `and`; a comma means **or**; `not` negates a whole query (or a condition in Level 4 syntax: `not (hover: hover)`); `only` is a legacy guard for very old browsers — harmless, no longer needed.

Media features you'll actually use

Feature	Values / example	Use
`width` (+ `min-`/`max-`)	`(width >= 40em)`	Layout breakpoints (the viewport width)
`height`	`(height < 500px)`	Short landscape phones: shrink sticky headers
`orientation`	`portrait` | `landscape`	Rarely needed — width usually says more
`aspect-ratio`	`(aspect-ratio >= 16/9)`	Full-screen media layouts
`hover` / `any-hover`	`hover` | `none`	Only show hover-reveal UI when the *primary* (or *any*) pointer (mouse, finger, pen) can hover
`pointer` / `any-pointer`	`fine` | `coarse` | `none`	Larger tap targets (the area you can touch to activate something) for touch (`coarse`)
`resolution`	`(resolution >= 2x)`	High-DPR (device pixel ratio, explained in §10.2) assets (prefer `srcset`/`image-set()`)
`prefers-color-scheme`	`light` | `dark`	Dark mode (10.5)
`prefers-reduced-motion`	`reduce` | `no-preference`	Tone down animation (10.5, Chapter 12)
`prefers-contrast`	`more` | `less` | `custom` | `no-preference`	Stronger borders/colours
`forced-colors`	`active` | `none`	Windows High Contrast / contrast themes
`print` (type)	`@media print`	Hide nav, show link URLs, black on white

Range syntax (Media Queries Level 4)

The old style uses `min-`/`max-` prefixes; the new **range syntax** uses comparison operators. It is Baseline since March 2023 and *widely available* since September 2025 (Chrome 104, Firefox 102, Safari 16.4).

Old	Range syntax
`(min-width: 640px)`	`(width >= 640px)`
`(max-width: 639.98px)`	`(width < 640px)`
`(min-width: 640px) and (max-width: 1023.98px)`	`(640px <= width < 1024px)`

Why the `.98`? With `max-width: 639px` and `min-width: 640px`, a window 639.5px wide (possible with zoom or fractional DPR) matches **neither**. Range syntax's `<` removes that gap.

`em` or `px` for breakpoints?

In media queries, `em` and `rem` are always relative to the **browser's default font size** (normally 16px), *not* your `html { font-size }`. So `40em` = 640px at defaults, but if a user sets 20px as their default, `40em` becomes 800px and your layout switches to one column earlier — exactly what a user with large text needs. Both units are fine; `em` ties breakpoints to text size.

Choosing breakpoints

- Let the **content** decide: widen the window until the layout looks wrong, and add a breakpoint *there*. Device-based lists go stale.
- For reference, popular frameworks (ready-made CSS libraries) use: Tailwind `sm 640 · md 768 · lg 1024 · xl 1280 · 2xl 1536`, Bootstrap 5 `576 · 768 · 992 · 1200 · 1400`.
- Fewer is better: two or three layout breakpoints, plus intrinsic techniques (10.9) and container queries (10.6) for components.

Media queries in JavaScript

`window.matchMedia(query)` returns a `MediaQueryList` (an object describing that query) with `.matches` and a `change` event (a notification your code can listen for) — useful for behaviour that CSS can't express (swap a component, pause a carousel — a rotating slideshow of items). Don't listen to `resize` and compare `innerWidth` yourself; `change` fires only when the answer flips.

```
const desktop = matchMedia('(width >= 64em)');
desktop.addEventListener('change', e => console.log(e.matches ? 'desktop' : 'not desktop'));
```

The last line of 10.4's output shows this firing: resizing 400 → 900 → 1100 → 1300 → 700 fired exactly **two** events. (`addListener()` is the deprecated older form.)

Common mistakes

- Desktop-first `max-width` queries mixed with mobile-first `min-width` ones — overlapping rules and gaps. Pick one direction.
- Putting a media query **before** the base rule it should override: equal specificity, so the later base rule wins.
- Using `(hover: hover)` alone to guess "desktop": laptops with touchscreens and phones with mice exist. Design for both.
- Breakpoints for specific devices ("iPhone 15") instead of the content.

Self-check

1. Write "between 600px and 900px, excluding 900" in range syntax. → `@media (600px <= width < 900px)`.
2. Does `em` in a media query follow `html { font-size: 62.5% }`? → No — it uses the browser's initial font size.
3. How does JS find out when a breakpoint is crossed? → `matchMedia(q).addEventListener('change', …)`.

10.4 Mobile-first in practice

The page

`code/02_mobile_first.html` is a small travel page. The base styles (no query) are the phone layout; two `min-width`-style queries add columns. The red badge at the top shows which rule set is active (it is a `::after` pseudo-element — a CSS-generated box — whose `content` each query overrides).

```
/* Base = phone: single column; normal flow + grid gap does the stacking */
.layout { display: grid; gap: 14px; padding: 14px; }
.cards  { display: grid; gap: 10px; }

@media (width >= 40em) {                    /* 640px: tablets, small laptops */
  header { display: flex; justify-content: space-between; align-items: center; }
  .cards { grid-template-columns: repeat(2, 1fr); }
}
@media (width >= 64em) {                    /* 1024px: sidebar appears */
  .layout { grid-template-columns: 1fr 240px; max-width: 1200px; margin-inline: auto; }
  .cards  { grid-template-columns: repeat(3, 1fr); }
}
```

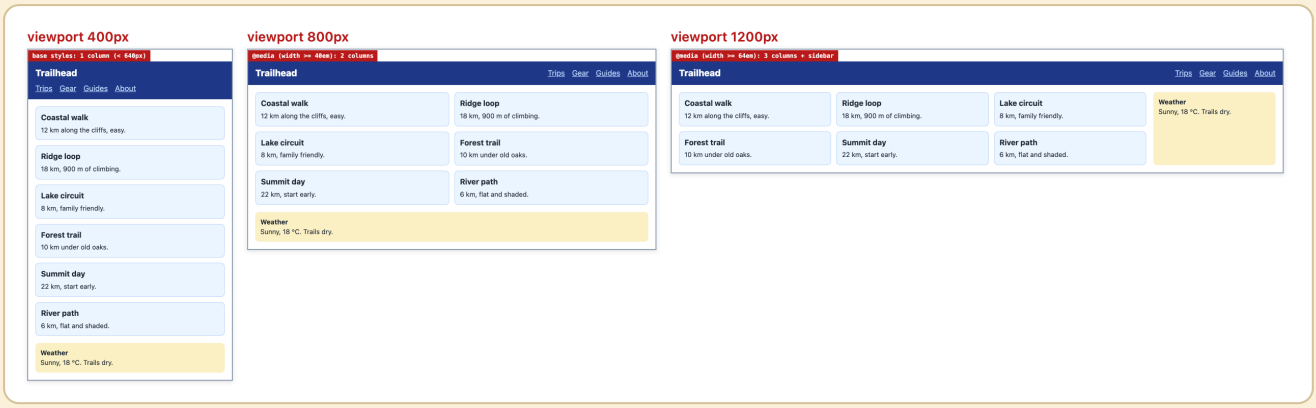


Figure 10.2 — One page, three viewport widths. Nothing in the HTML changes; only which rules apply.

Output of `code/02_media_queries.mjs` — which queries match and what the browser resolved the card grid to:

```
 400px  (width >= 40em) no   (width >= 64em) no   cards: 372px
 800px  (width >= 40em) yes  (width >= 64em) no   cards: 381px 381px
1200px  (width >= 40em) yes  (width >= 64em) yes  cards: 299.328px 299.328px 299.344px
matchMedia events: [ 'change -> desktop at 1100px', 'change -> not desktop at 700px' ]
```

Check by hand at 1200px: `.layout` is `max-width: 1200px` minus 2 × 14px padding = 1172px; minus the 240px sidebar and a 14px gap = 918px for `.cards`; minus two 10px gaps = 898px; ÷ 3 = **299.33px** per card column. ✓

Why it works

- **Normal flow** is already mobile-first: block elements stack. The base CSS mostly styles, it barely lays out.
- Each query **adds** a layout; nothing has to be "undone".
- The **DOM order** (header, cards, aside — the order in the HTML) is the reading order at every width, so keyboard and screen-reader (software that reads pages aloud) users get the same sequence (Chapter 3). Reordering visually with `order` or grid placement at a breakpoint breaks that — avoid it for meaningful content.

Responsive navigation (common pattern)

At phone width a long nav usually becomes a **menu button** that shows/hides the list. Accessible recipe: a real `<button aria-expanded="false" aria-controls="menu">` (ARIA attributes that tell screen readers whether the menu is open and which element it controls) that toggles the list (a few lines of JS), or the

Popover API (`popover` attribute, Baseline since January 2025) for a no-JS disclosure (show/hide widget). The flex-wrap nav in the demo is the simplest option when there are only a few links.

Responsive tables

Wide data tables can't reflow into one column without losing their meaning. Wrap them in a scroll container (a box that scrolls its own content): `<div class="table-wrap" tabindex="0" role="region" aria-label="Prices">` with `.table-wrap { overflow-x: auto; }` — WCAG allows two-dimensional scrolling for data tables (Chapter 5).

Self-check

1. At 800px, which of the two queries match? → Only `(width >= 40em)`.
2. Why write the queries after the base rules? → Same specificity, so the later rule wins the cascade.

10.5 User-preference media queries

What it is

Operating systems let people choose a dark theme, reduce motion, or increase contrast. The **user-preference media features** (Media Queries Level 5) expose those choices to CSS. All of the following are Baseline:

Feature	Baseline	Typical use
`prefers-color-scheme: dark`	widely available (2020)	Dark theme
`prefers-reduced-motion: reduce`	widely available (2020)	Remove or soften animation — full treatment in Chapter 12
`prefers-contrast: more`	widely available (2022)	Solid borders, stronger colours
`forced-colors: active`	widely available (2022)	Windows contrast themes replace your colours; restore borders on custom controls, use system colours like `CanvasText`
`prefers-reduced-transparency`	**not** Baseline (Chromium only)	Remove frosted-glass effects
`prefers-reduced-data`	not supported anywhere yet	—

Dark mode — the essentials

1. **Declare support**: `<meta name="color-scheme" content="light dark">` in the head and/or `:root { color-scheme: light dark; }`. The browser then themes its **own** UI (user interface) — default page background, form controls, scrollbars — and avoids a white flash in dark mode. (`color-scheme`: Baseline 2022.)
2. **Colour tokens** (named colour values such as `--bg`) as custom properties (CSS variables, §6.9), light values first.
3. **Override the tokens** inside `@media (prefers-color-scheme: dark)`. Components only ever use the tokens.
4. For one-off values, `light-dark(lightValue, darkValue)` picks based on the element's **used colour scheme** (light or dark, as finally decided by `color-scheme` and the user's setting) — no media query. It needs `color-scheme: light dark` to be set, and it is Baseline since May 2024 (newly available).

```
:root { color-scheme: light dark;
        --bg: #ffffff; --text: #1f2937; --card: #eff6ff; --accent: #1d4ed8; --border: #bfdbfe; }
@media (prefers-color-scheme: dark) {
  :root { --bg: #0f172a; --text: #e2e8f0; --card: #1e293b; --accent: #93c5fd; --border: #334155; }
}
@media (prefers-contrast: more) { :root { --border: currentColor; } }

body  { background: var(--bg); color: var(--text); }
.note { color: light-dark(#92400e, #fcd34d); }
```

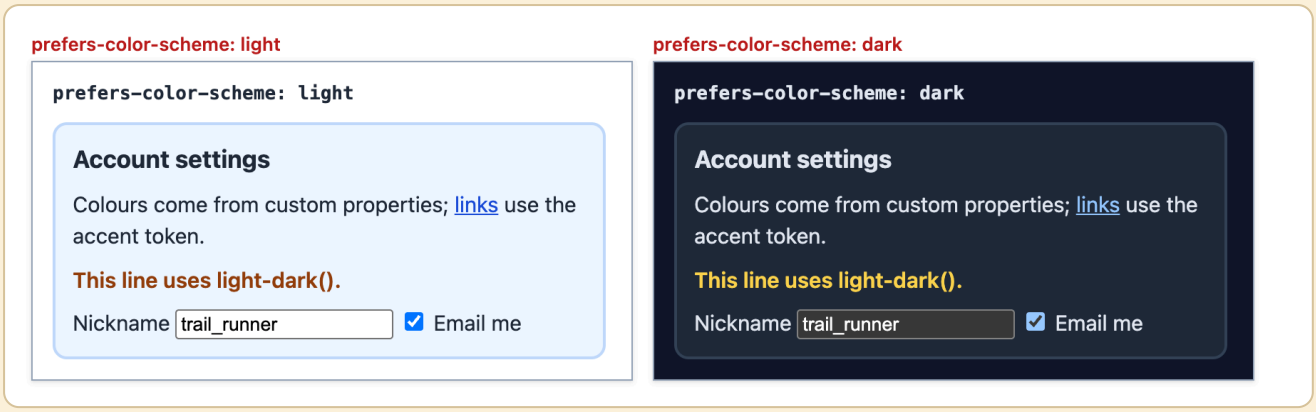


Figure 10.3 — `code/03_preferences.html` under emulated `prefers-color-scheme: light` and `dark`. Notice the text input and checkbox: nobody styled them — `color-scheme` made the browser draw its dark versions. The dark screenshot alone can be reproduced with `node tools_web_shot.mjs chapters/10-responsive-design/code/03_preferences.html out.png 420 --dark`.

`code/03_preferences.mjs` reads the computed colours under three emulated settings:

```
light (default)        bg rgb(255, 255, 255)  text rgb(31, 41, 55)  card border rgb(191, 219, 254)  note rgb(146, 64, 14)  <input> bg rgb(255, 255, 255)
dark                   bg rgb(15, 23, 42)  text rgb(226, 232, 240)  card border rgb(51, 65, 85)  note rgb(252, 211, 77)  <input> bg rgb(59, 59, 59)
light + more contrast  bg rgb(255, 255, 255)  text rgb(31, 41, 55)  card border rgb(31, 41, 55)  note rgb(146, 64, 14)  <input> bg rgb(255, 255, 255)
```

With more contrast, the card border became the text colour (`currentColor`). The input background `rgb(59, 59, 59)` in dark mode is Chrome's built-in dark control colour.

A practical note from making these figures: headless Chrome follows the **computer's** theme when nothing is emulated. On a Mac in dark mode, a page with `color-scheme: light dark` screenshots dark unless you emulate `light` explicitly — `measure.mjs` does exactly that.

Letting users override

Many sites add a theme toggle that sets `data-theme="dark"` on `<html>` and stores the choice. Then tokens are defined for `[data-theme="dark"]` too, and the media query applies only when no choice was made (`:root:not([data-theme="light"])`). That architecture — plus `color-scheme` per theme — is covered in Chapter 13.

Designing a dark theme well

- Don't invert: use dark greys/navies (pure `#000` with pure white text causes halation — a glowing, blurry halo around letters — for many readers), lighten accent colours, and re-check contrast ≥ 4.5:1 (Chapter 11).
- Shadows barely show on dark backgrounds — express elevation (how "raised" a surface looks) with slightly lighter surfaces.
- Images: dim very bright photos slightly (`filter: brightness(.9)`) or supply dark versions with `<picture><source media="(prefers-color-scheme: dark)" …>`.

Reduced motion in one rule

```
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after { animation-duration: 0.01ms !important; animation-iteration-count: 1 !important;
                           transition-duration: 0.01ms !important; scroll-behavior: auto !important; }
}
```

This blunt reset is common; §12.9 explains a gentler "motion only when allowed" pattern and what WCAG requires.

Self-check

1. What does `color-scheme: light dark` change even if you write no other dark CSS? → The browser's defaults: canvas background, text colour, form controls, scrollbars.
2. When does `light-dark()` pick its second value? → When the element's used colour scheme is dark (requires `color-scheme` to allow dark).
3. What should a site do when `forced-colors: active`? → Let the system colours win; make sure custom controls still have visible borders/focus indicators (e.g. with system colours like `ButtonText`), and don't rely on background images for meaning.

10.6 Container queries

What it is

A **container query** (`@container`) styles an element based on the size of an **ancestor container** (an element somewhere above it in the HTML tree) instead of the viewport. It is Baseline since February 2023 and **widely available** since August 2025 (Chrome 105, Firefox 110, Safari 16).

The problem it solves

Components are reused in different places: a product card sits in a wide main column, a narrow sidebar and a two-up grid — all on the same screen. A media query only knows the *viewport* width, so it can't tell those three cards apart. You used to add modifier classes (extra classes that switch a component's variant, e.g. `.card--compact`) by hand. With container queries the card adapts to the space **it is given**, wherever it is placed.

Analogy: media queries ask "how big is the room?"; container queries ask "how big is the box I'm sitting in?"

How it works

1. Mark an ancestor as a **size container**: `container-type: inline-size` (width only — the usual choice) or `size` (both axes, but then the container can't get its height from its content, so you must give it one). The shorthand `container: name / type` also sets an optional **name**.
2. Write rules inside `@container [name] (condition) { … }`. They apply to **descendants** of the matching container.
3. The browser finds, for each element, the **nearest ancestor** container (with that name, if given) and evaluates the condition against **its** content-box size.
4. A container can't style *itself* with its own query — that would be circular (its size would depend on styles that depend on its size). This is why the demo puts the container on a wrapper (`.card-slot`) and styles the `.product` inside.

```
.card-slot { container: card / inline-size; }       /* name "card", measured inline */

.product    { display: grid; gap: 8px; }             /* base: picture above text */
.product h3 { font-size: clamp(15px, 3cqi + 8px, 24px); }

@container card (width >= 400px) { .product { grid-template-columns: 160px 1fr; } }
@container card (width >= 600px) { .product { grid-template-columns: 240px 1fr; padding: 16px; } }
```

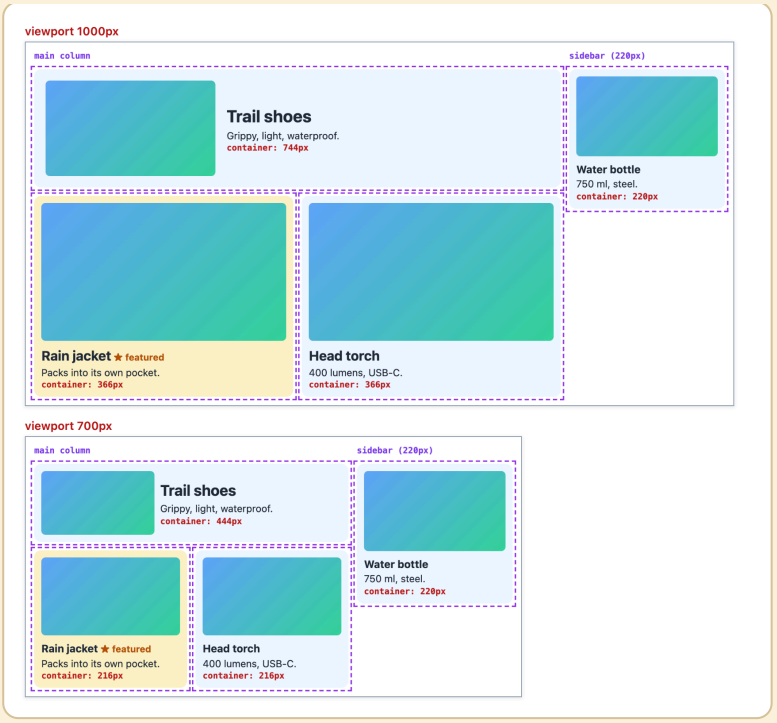


Figure 10.4 — `code/04_container_queries.html`. Dashed purple = the containers; each card prints its container width in red. Same component, same viewport, three different layouts.

Output of `code/04_container_queries.mjs`:

```
viewport 1000px
  Trail shoes   container 744px  columns 240px 456px    h3 24px      ::after none
  Rain jacket   container 366px  columns 346px          h3 18.98px   ::after " ★ featured"
  Head torch    container 366px  columns 346px          h3 18.98px   ::after none
  Water bottle  container 220px  columns 200px          h3 15px      ::after none
viewport 700px
  Trail shoes   container 444px  columns 160px 256px    h3 21.32px   ::after none
  Rain jacket   container 216px  columns 196px          h3 15px      ::after " ★ featured"
  Head torch    container 216px  columns 196px          h3 15px      ::after none
  Water bottle  container 220px  columns 200px          h3 15px      ::after none
```

The sidebar card (220px container) looks identical at both viewport widths, while the "Trail shoes" card switches from the ≥600px layout to the ≥400px layout. A media query could not do this.

Container query units

Inside a container you can size things relative to it:

Unit	1% of the query container's…
`cqw` / `cqh`	width / height
`cqi` / `cqb`	inline size / block size (writing-mode aware — prefer these)
`cqmin` / `cqmax`	smaller / larger of the two

If no suitable container exists, they fall back to the **small viewport** units (`svw`/`svh`…). Check the heading sizes: `clamp(15px, 3cqi + 8px, 24px)` in a 366px container = 3 × 3.66 + 8 = **18.98px** ✓; in a 220px container = 14.6px → clamped up to **15px** ✓; in 744px = 30.3px → capped at **24px** ✓.

Style queries

`@container style(--featured: yes)` tests a **custom property's value** on the container instead of its size. Every element is a style container by default, so no `container-type` is needed. In the demo, `.featured { --featured: yes; }` turns the "Rain jacket" card yellow with a star — a design-token (named design value) switch without a modifier class on the component. Status: **Baseline since May 2026** (newly available:

Chrome 111, Safari 18, Firefox 151), and only for **custom properties**; querying ordinary properties (e.g. `style(display: flex)`) is not supported. Treat it as an enhancement for a while.

Other container queries (not Baseline)

`scroll-state()` queries — "is this sticky header stuck?", "is this carousel item snapped?" — are Chromium-only (only browsers built on Google's engine, Chrome 133+). Leave them out of anything that must work everywhere.

Container queries vs media queries

Use a **media query** for…	Use a **container query** for…
Page-level layout (how many columns the page has)	Components that appear in different-sized slots
User preferences (`prefers-*`), print, pointer type	Adapting a card/widget's internal layout
Anything about the device	Sizing text/spacing relative to the component (`cqi`)

They work together: media queries shape the page, container queries let each piece adapt to the space the page gave it.

Common mistakes

- Putting `container-type` on the element you want to restyle — it can only restyle its **descendants**.
- `container-type: size` on an element without an explicit height: it collapses to 0 height, because size containment (the browser treating the box as if it were empty when sizing it) ignores its content.
- Forgetting that `inline-size` containment means the container's **width** can't depend on its children — a `container-type: inline-size` element inside a shrink-to-fit parent (one whose width comes from its content: a flex item with `flex: none`, an inline-block, a float) may collapse. Give it a width from outside (grid track, block flow, `flex: 1`).
- Naming: an unnamed `@container (width > 400px)` matches the **nearest** size container of any name, which might not be the one you meant. Name containers in design systems (a team's shared library of components).

Fallback for older browsers

```
@supports not (container-type: inline-size) { .product { /* a sensible single layout */ } }
```

`@supports (feature: value)` is a **feature query**: its rules apply only if the browser understands that declaration. Container queries are widely available now, so this is only needed for old devices.

In the interview

30-second answer: "Media queries respond to the viewport; container queries respond to the size of an ancestor that I've declared a container with `container-type: inline-size`. That makes components portable: the same card can be stacked in a sidebar and side-by-side in the main column on the same screen. The container can't query itself, so I put it on a wrapper, and I use `cqi` units for sizes relative to it. Size queries are Baseline since 2023; style queries on custom properties became Baseline in 2026."

Follow-ups:

- *Why can't the container use `container-type: size` by default?* → Size containment means its size can't depend on its content; for height that would usually collapse it to 0. `inline-size` only fixes the width dependency.
- *What happens to `cqi` if there's no container?* → It falls back to small-viewport units (`svi`).
- *Would you drop media queries?* → No: page layout and user preferences still belong to media queries.

Self-check

1. Where does `container-type` go: on the card or its wrapper? → On the wrapper (an ancestor); the card's styles come from queries about the wrapper.
2. A container is 500px. What is `10cqi`? → 50px.
3. What does `@container style(--theme: dark)` test? → Whether the nearest container's computed `--theme` custom property equals `dark`.

10.7 Fluid typography with `clamp()`

What it is

Fluid typography makes font sizes (and often spacing) change **smoothly** with the viewport between a minimum and a maximum, instead of jumping at breakpoints. The tool is `clamp(MIN, PREFERRED, MAX)` — Baseline since 2020 (now widely available), along with `min()` and `max()`:

- `clamp(MIN, VAL, MAX)` = `max(MIN, min(VAL, MAX))`: use `VAL`, but never below `MIN` or above `MAX`.

How to derive the formula

Goal: body text **16px at a 320px viewport** growing to **24px at a 1200px viewport**.

1. **Slope** = change in size ÷ change in viewport = (24 − 16) / (1200 − 320) = 8 / 880 = 0.00909. Since `1vw` = 1% of the viewport, that is **0.909vw**.
2. **Intercept** (the size at a 0px viewport) = 16 − 0.00909 × 320 = 13.09px = **0.818rem**.
3. Preferred value = `0.818rem + 0.909vw`, clamped to `1rem … 1.5rem`:

```
:root {
  --step-0: clamp(1rem, 0.818rem + 0.909vw, 1.5rem);      /* 16px → 24px  */
  --step-3: clamp(1.75rem, 1.296rem + 2.273vw, 3rem);     /* 28px → 48px  */
  --space:  clamp(0.75rem, 0.341rem + 2.045vw, 1.875rem); /* 12px → 30px  */
}
p  { font-size: var(--step-0); }
h1 { font-size: var(--step-3); }
```

`code/05_fluid_type.mjs` compares Chrome's computed sizes with the straight-line formula:

```
slope = 0.909vw, intercept = 13.091px = 0.818rem
  280px  p 16.00px (by hand 16.00)   h1 28.00px
  320px  p 16.00px (by hand 16.00)   h1 28.01px
  600px  p 18.54px (by hand 18.55)   h1 34.37px
  900px  p 21.27px (by hand 21.27)   h1 41.19px
 1200px  p 24.00px (by hand 24.00)   h1 48.00px
 1600px  p 24.00px (by hand 24.00)   h1 48.00px
default font 20px, viewport 1000px, rem + vw (ours)  -> 25.45px
default font 20px, viewport 1000px, vw only (bad)    -> 16px
```

The 0.01px differences come from rounding the CSS coefficients to three decimals.

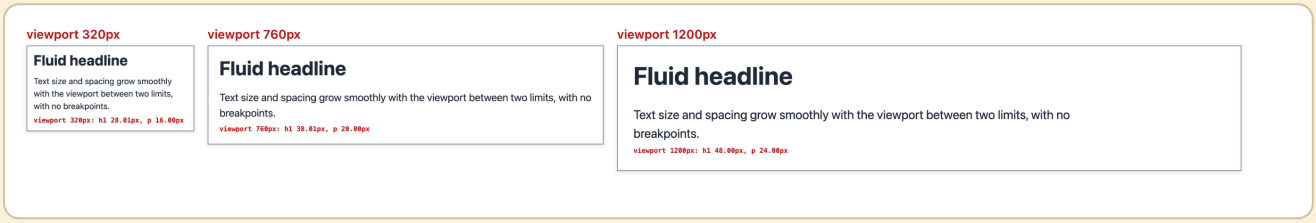


Figure 10.5 — `code/05_fluid_type.html` at three widths. Each page prints its own computed sizes.

Why the `rem` part is essential (accessibility)

The last two output lines emulate a user who changed the browser's **default font size** to 20px. The `rem + vw` formula grows to 25.45px (0.818 × 20 + 9.09). A pure `vw` size stays at **16px** — it ignores the user's setting, and it also doesn't grow with **page zoom** in the way users expect. WCAG 2.2 SC 1.4.4 (Resize

text) requires text to scale to 200%; keep a `rem` term in the preferred value, keep `MIN` and `MAX` in `rem`, and keep the ratio MAX/MIN at or below about 2.5 so that zooming still enlarges text meaningfully.

Type scales

Designers usually pick a **modular scale** (a set of font sizes related by one multiplier) — each step = previous × ratio (e.g. 1.2 "minor third" on small screens, 1.333 "perfect fourth" on large). Fluid type interpolates (fills in the in-between values along a straight line for) **each step** between its small-screen and large-screen size, which is what `--step-0 … --step-5` token sets from generators such as Utopia (a free online fluid-type calculator) do. The same interpolation works for spacing (`--space`), making padding and gaps breathe with the screen.

Common mistakes

- `font-size: 4vw` alone — unreadable on phones, huge on monitors, ignores user settings.
- Mixing units in the wrong places: `clamp(16px, …, 24px)` ignores the user's default for the limits; prefer `rem`.
- Forgetting `line-height`: large headings need a smaller unitless line-height (a plain number such as 1.2, meaning 1.2 × the font size) (1.1–1.2), body text ~1.5 (Chapter 11).

Self-check

1. Derive the preferred value for 18px at 400px → 22px at 1400px. → Slope 4/1000 = 0.4vw; intercept 18 − 1.6 = 16.4px = 1.025rem → `clamp(1.125rem, 1.025rem + 0.4vw, 1.375rem)`.
2. Why not `clamp(1rem, 2vw, 1.5rem)`? → It works, but the middle term has no `rem` part, so between the limits it ignores the user's font-size setting.

10.8 Responsive images recap (and `image-set()`)

Chapter 4 §4.3 covers responsive images in depth. The short version:

Need	Tool
Image never overflows its container	`img { max-width: 100%; height: auto; }` + `width`/`height` attributes (reserves space → no layout shift, i.e. content doesn't jump when the image arrives)
Same picture, different file sizes; browser picks by layout width and DPR	`<img srcset="a-400.jpg 400w, a-800.jpg 800w" sizes="(width >= 64em) 33vw, 100vw">`
Same size on screen, sharper on high-DPR screens	`srcset="logo.png 1x, logo@2x.png 2x"`
Different crop at different widths (**art direction**)	`<picture><source media="(width >= 800px)" srcset="wide.jpg"><img src="square.jpg" alt="…"></picture>`
Newer format with fallback	`<picture><source type="image/avif" srcset="a.avif"><img src="a.jpg" alt="…"></picture>`
Keep a box's shape before content loads	`aspect-ratio: 16 / 9`

`image-set()` for CSS background images

`srcset` only works on `<img>`. For **background images**, CSS has `image-set()` — Baseline since September 2023, widely available since March 2026 (older Safari needed `-webkit-image-set()`):

```
.hero {
  background-image: url("media/hills-400.jpg");                  /* fallback */
  background-image: image-set("media/hills-400.jpg" 1x,
                              "media/hills-800.jpg" 2x);
  background-size: cover;
}
```

Candidates can also be chosen by format, e.g. `image-set("a.avif" type("image/avif"), "a.jpg" type("image/jpeg"))`. `code/06_image_set.mjs` loads `code/06_image_set.html` at three pixel ratios and logs the network:

```
devicePixelRatio 1: downloaded hills-400.jpg
devicePixelRatio 2: downloaded hills-800.jpg
devicePixelRatio 3: downloaded hills-800.jpg
```

Only **one** file is downloaded per screen — the whole point. A background image is decoration: if the picture carries meaning, use an `<img>` with `alt` (the demo adds `role="img"` and `aria-label` only because it is a demo of a decorative technique).

10.9 Intrinsic layouts — responsive without breakpoints

What it is

Intrinsic design (a term from Jen Simmons, a CSS designer and advocate) means letting the layout algorithms (the browser's built-in sizing rules) respond to the available space themselves, using `min()`, `max()`, `clamp()`, `flex-wrap`, `auto-fit` and `minmax()`. Fewer breakpoints means fewer "in-between" widths where things look odd, and components that work in any container.

Three patterns

`code/07_intrinsic.html` — no `@media` anywhere:

```
/* 1. Readable centred column: at most 60ch, and never touching the screen edges */
.wrapper { width: min(100% - 2rem, 60ch); margin-inline: auto; }

/* 2. Sidebar that wraps when the content would get narrower than 60% */
.with-sidebar          { display: flex; flex-wrap: wrap; gap: 10px; }
.with-sidebar > .side  { flex: 1 1 150px; }                    /* wants ~150px */
.with-sidebar > .content { flex: 999 1 0; min-inline-size: 60%; } /* takes the rest */

/* 3. Card grid: as many ≥140px columns as fit (Chapter 9 §9.3) */
.grid { display: grid; gap: 10px; grid-template-columns: repeat(auto-fit, minmax(min(140px, 100%), 1fr)); }
```

How pattern 2 works: the content's huge `flex-grow` (999) means it takes almost all free space, so the sidebar stays near its 150px basis. When the line can't hold `150px + gap + 60%`, flex-wrap moves the sidebar to its own line, where it grows to full width. That is a "breakpoint" chosen by the content, not the viewport — and it works inside any container.

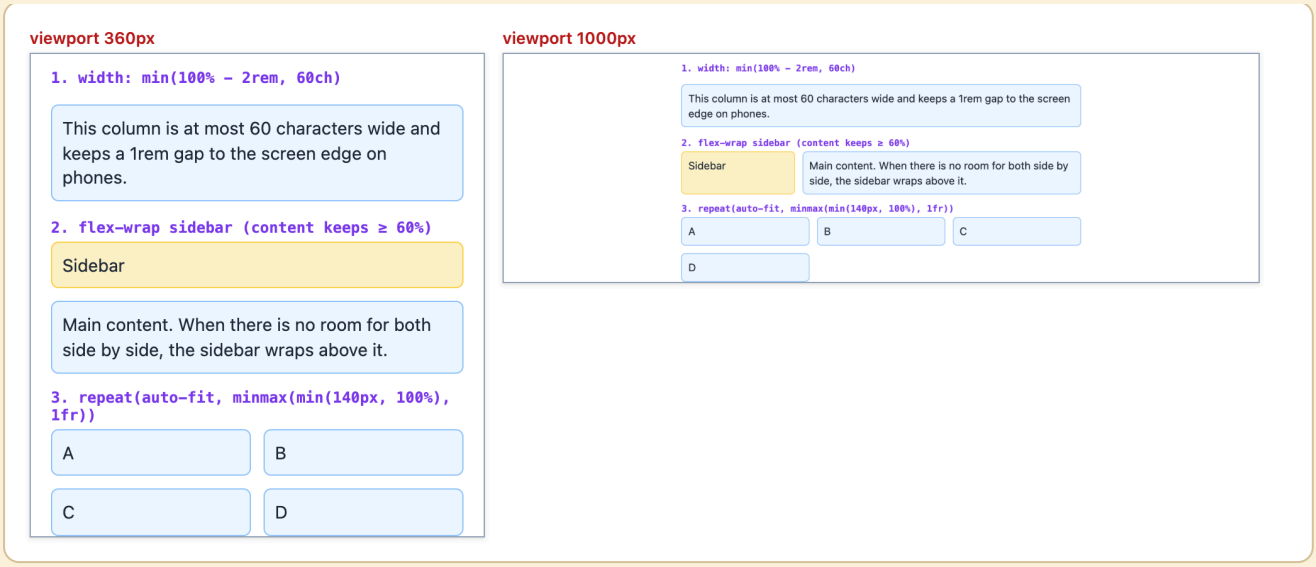


Figure 10.6 — The same page at 360px and 1000px.

Output of `code/07_intrinsic.mjs`:

```
 360px  wrapper 328.0px | sidebar wrapped above content | grid 2 column(s)
 480px  wrapper 448.0px | sidebar beside content (150.3px + 287.7px) | grid 3 column(s)
1000px  wrapper 529.1px | sidebar beside content (150.3px + 368.8px) | grid 3 column(s)
```

By hand at 480px: wrapper = min(480 − 32, 60ch ≈ 529px) = 448px. The line needs 150 + 10 + 0.6 × 448 = 428.8px ≤ 448, so the sidebar stays beside; free space 448 − 10 − 150 = 288px is shared 1 : 999, so the sidebar gets 150 + 0.288 = 150.3px ✓. At 360px the wrapper is 328px and the line would need 150 + 10 + 196.8 = 356.8px > 328, so it wraps ✓.

Other intrinsic tools

- `flex-wrap: wrap` + `flex: 1 1 12rem` on items — a row of buttons that becomes a column when needed (Chapter 8).
- `max-inline-size: 65ch` on text — the **measure** (line length) that reads best (Chapter 11).
- `aspect-ratio` for media placeholders; `object-fit: cover` to crop images into them (Chapter 11).

When not to use it

When the design genuinely needs a different structure (e.g. a sidebar that becomes a bottom tab bar), a media query states that intent more clearly than a clever flex trick.

10.10 Testing responsive design

What to test

Check	How	Pass criterion
No sideways scrolling	Resize from 320px to 1920px+	`scrollWidth ≤ clientWidth` at every width (except data tables, maps, etc.)
Reflow (WCAG 2.2 SC 1.4.10, AA)	View at 320 CSS px wide (= 1280px at 400% zoom)	All content and functions available with vertical scrolling only
Resize text (SC 1.4.4, AA)	Browser zoom 200%; also raise the default font size	No clipped or overlapping text, nothing lost
Text spacing (SC 1.4.12, AA)	Apply line-height 1.5, paragraph spacing 2em, letter 0.12em, word 0.16em	Nothing clipped — avoid fixed heights on text boxes
Target size (SC 2.5.8, AA, new in 2.2)	Inspect tap targets	At least 24 × 24 CSS px (or enough spacing)
Orientation (SC 1.3.4)	Rotate the device	Works in both, unless essential
Touch vs mouse	Real phone; `(hover: none)`	Nothing is reachable *only* on hover
Preferences	Emulate dark, reduced motion, forced colours	Readable, no lost UI

Tools

- **DevTools device mode**: Chrome/Edge (Ctrl/Cmd+Shift+M), Firefox Responsive Design Mode (Ctrl/Cmd+Shift+M), Safari Responsive Design Mode (Develop menu). Drag the width handle slowly — you'll spot the widths where things break. Chrome's Rendering panel emulates `prefers-color-scheme`, `prefers-reduced-motion`, `forced-colors` and more; Elements shows a `container` badge and lets you inspect container queries.
- **Real devices**: emulators (software imitations of a device) don't reproduce touch precision, the dynamic toolbar (`dvh`), the on-screen keyboard or real CPU (processor) speed. Test at least one Android phone and one iPhone; remote debugging (inspecting a page running on a phone from your computer — Chrome `chrome://inspect`, Safari Web Inspector) attaches DevTools to them.
- **Automated screenshots and checks** in CI (continuous integration — automatic checks run on every code change) with Puppeteer or Playwright (browser-automation libraries), like this chapter's scripts.

An automated overflow test

`code/08_overflow_check.mjs` opens pages at 320/768/1280px and reports horizontal overflow and the elements sticking out:

```
const check = page => page.evaluate(() => {
  const vw = document.documentElement.clientWidth;
  const culprits = [...document.body.querySelectorAll('*')]
    .filter(el => el.getBoundingClientRect().right > vw + 0.5 || /* … content overflowing its box … */)
    .map(el => el.tagName.toLowerCase() + (el.className ? '.' + el.className : ''));
  return { overflow: document.documentElement.scrollWidth > vw, scrollWidth: document.documentElement.scrollWidth,
culprits };
});
```

```
02_mobile_first.html   320px  ok
02_mobile_first.html   768px  ok
02_mobile_first.html  1280px  ok
08_broken.html         320px  OVERFLOW (page is 575px wide): div.promo, img, p
08_broken.html         768px  ok
08_broken.html        1280px  ok
08_fixed.html          320px  ok
08_fixed.html          768px  ok
08_fixed.html         1280px  ok
```

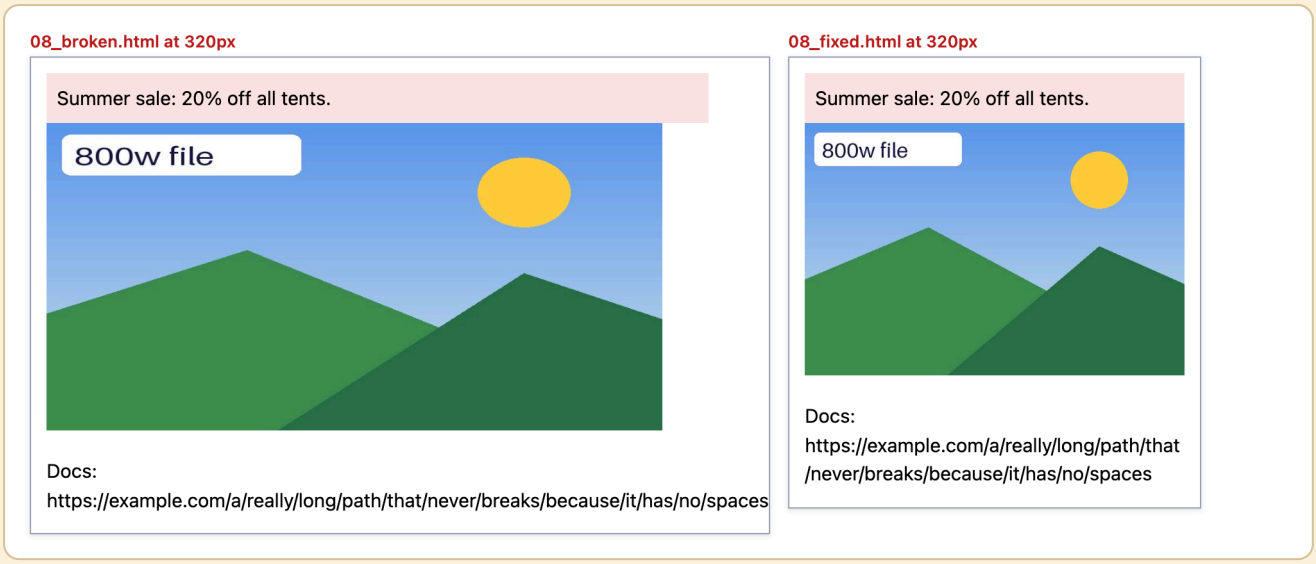


Figure 10.7 — The three most common causes of sideways scrolling on phones, and their one-line fixes.

Bug in `08_broken.html`	Fix in `08_fixed.html`
`.promo { width: 500px }`	`max-width: 500px` — fixed widths are the #1 cause
`<img width="480">` with no CSS	`img { max-width: 100%; height: auto; }`
Long URL with no break opportunities	`overflow-wrap: anywhere` (or `break-word`) on text containers

Other frequent culprits: `100vw` widths (10.2), grid columns with `1fr` that can't shrink below a long word (`minmax(0, 1fr)`, §9.10), flex items with `min-width: auto` (Chapter 8), `<pre>` blocks and wide tables (wrap them in `overflow-x: auto`).

In the interview

30-second answer: "I build mobile-first: the base CSS is the single-column phone layout and `min-width` media queries add columns where the content needs them. The viewport meta tag makes phones lay out at device width. I use intrinsic techniques — flex-wrap, `repeat(auto-fit, minmax())`, `clamp()` for fluid type with a `rem` term — so I need few breakpoints, and container queries for components that live in different-sized slots. I respect user preferences: `prefers-color-scheme` with `color-scheme`, `prefers-reduced-motion`. Images use `srcset`/`sizes`. I test in device mode, on real phones, at 320px and 200% zoom for WCAG reflow."

Follow-ups:

- *Why does `100vh` misbehave on mobile?* → It equals the large viewport (toolbars hidden), so with toolbars visible the bottom is cut off. Use `svh` or `dvh`.
- *How do you pick breakpoints?* → Where the content breaks, not device widths; prefer em so they follow the user's font size.
- *Media vs container queries?* → Viewport vs ancestor size; page layout vs component layout.
- *How do you avoid horizontal scrolling?* → No fixed widths (use `max-width`), flexible media, `overflow-wrap`, `minmax(0,1fr)`, and an automated check at 320px.

Self-check

1. What width does WCAG's Reflow criterion test? → 320 CSS px (equivalent to 1280px at 400% zoom).
2. Name three causes of horizontal overflow. → Fixed widths, images without `max-width`, long unbreakable strings (also `100vw`, rigid grid tracks, wide tables).
3. What can't DevTools device mode emulate well? → Real touch, dynamic toolbars/`dvh`, on-screen keyboard, real performance.

Chapter 10 TL;DR

- **Responsive** = fluid layout + flexible media + conditional CSS. Build **mobile-first**: base CSS is the phone layout; `(width >= …)` queries add columns.
- `<meta name="viewport" content="width=device-width, initial-scale=1">` — without it phones lay out at ~980px and shrink (measured scale 0.408 on a 400px phone). Never disable zoom.
- **Viewport units**: `svh` (toolbars shown), `lvh` (hidden), `dvh` (live); `vh` ≈ `lvh` on phones; `100vw` includes the scrollbar.
- **Media queries**: type + features + `and` / `,` / `not`. Range syntax `(640px <= width < 1024px)` is widely available and removes boundary gaps. `em` in queries = browser default font size. JS: `matchMedia(q)` + `change` event.
- **Preferences**: `prefers-color-scheme` + `color-scheme: light dark` (themes form controls too) + token overrides; `light-dark()` (Baseline 2024); `prefers-contrast`, `forced-colors`, `prefers-reduced-motion`.
- **Container queries** (widely available since Aug 2025): `container: name / inline-size` on a wrapper, `@container name (width >= 400px)` for descendants, `cqi` units. Style queries on custom properties: Baseline 2026. Scroll-state queries: Chromium only.
- **Fluid type**: `clamp(MIN, a·rem + b·vw, MAX)`; slope = Δsize/Δviewport, intercept = size − slope × viewport; keep a `rem` term so user settings and zoom still work (vw-only stayed 16px when the default was 20px).
- **Images**: `max-width: 100%`, `srcset`/`sizes`, `<picture>`; `image-set()` for backgrounds (downloads one file per DPR).
- **Intrinsic layouts**: `min(100% - 2rem, 60ch)`, flex-wrap sidebar, `repeat(auto-fit, minmax(min(X,100%),1fr))` — breakpoints chosen by content.
- **Test**: 320px reflow, 200% zoom, text spacing, 24px targets, real devices, dark/reduced motion, automated overflow checks.

Next: Chapter 11 — Typography, Colour and Visual Effects.